


# Generative Alin life steward — AIVA

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## Abstract

Aiva is an advanced **AI-powered assistant** designed to enhance the efficiency and simplicity of daily life management through **adaptive decision support, optimized scheduling,** and **personalized health coaching.** By dynamically integrating real-time user data with environmental variables**,** Aiva effectively addresses the growing complexity of modern family coordination challenges. Unlike conventional life management tools, which are typically **fragmented** and dependent on **manual data entry,** Aiva leverages **generative artificial intelligence** and **peripheral technologies** such as cameras for **automated data collection** and **intelligent processing**, delivering a **seamless** and **highly efficient user experience**.

## Introduction

### Integrated solution, no fragmentation

Modern life management faces a **critical challenge**: traditional tools (e.g., generic calendar apps, static diet plans) fail to adapt to **dynamic personal needs, real-time scenarios, and holistic wellness goals.**

### Human-centric Computing: Integration and Personalization

This is essential because while individuals increasingly seek **personalized support** for **health, productivity, and decision-making**, existing solutions remain fragmented and lack contextual awareness. Aiva is an **AI-driven life assistant** that simplifies daily routines through **adaptive nutrition tracking, smart scheduling, and proactive decision-making**. By blending real-time data analysis with human-centric design, it replaces rigid tools with personalized support—**learning habits**, optimizing tasks, and empowering healthier, intentional choices.

## Methodology

Require Analysis → Design Alternatives → Prototype Development → Evaluating → Require Analysis

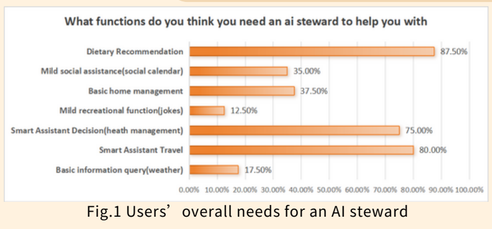



Fig.1 Users' overall needs for an AI steward




Fig.2 Users' specific needs

An online survey with both open-end and closed-end questions and received 70 valid results.

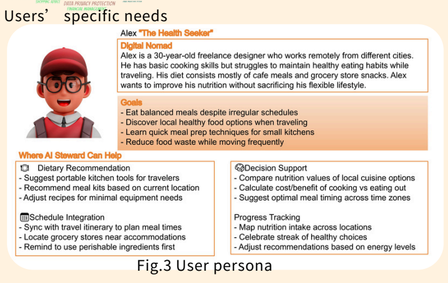



Fig.3 User persona

Based on the results of queries(Fig.1 and Fig.2) and interviews(Fig.3), the target users' requirements can be summarized as follows:

1. Dietary Recommendation Features Functional Diversity（87.50%）: Users expect diverse features to meet various dietary needs. Algorithm Accuracy: Users desire more precise algorithms for personalized recommendations.
2. Smart Assistant Decision Support Health Management（75.00%）: Users prioritize support for health management to better monitor their well-being. Financial/Shopping Management: Users also seek assistance in financial and shopping decisions to optimize daily choices.
3. Smart Assistant Travel Planning Schedule Management: Users want assistance in organizing daily schedules. Travel Route Planning: Users expect recommendations for optimal travel routes to enhance their travel experience.

### Plan A

Utilize AI to create a one-stop solution for life optimization by integrating users' needs in diet, decision-making, and scheduling.

**Advantage**

Comprehensive features, covering high frequency scenes of life, data interworking (such as diet affecting schedule).
users can have personalization plans.

**Disadvantage**

The operation is complex and it takes longer for users to learn

### Plan B

Use generative AI to provide recommendations for food and drink. Recommend suitable daily activities and provide travel planning

**Advantage**

Automatically optimize meeting schedules and travel itineraries, intelligently pull email to-do lists, and coordinate team schedules

**Disadvantage**

Users need to manually enter data to get the proposal, and the provision of data is relatively tedious

### Plan C

Use an integrated dialog box to implement the different functions, and make comments entirely through the user's question.

**Advantage**

It's easy to use, just ask questions.

**Disadvantage**

In long-term use, the requirements for different aspects may be recorded in a confused way, and the user experience is poor. And lack of pertinence, did not do private customization

### Prototype Development

#### Decision Support System

Utilizes AI to assist in making informed decisions, helping users navigate life or career-related dilemmas with greater clarity and confidence.

#### Smart Eating Assistant

Uses real-time camera analysis to identify the types and nutritional values of food consumed, providing detailed nutritional insights and analysis.

#### Scheduling System

Based on the imported schedule or class timetable, the system sets up reminders and provides relevant support information,
such as recommending nearby dining and accommodation options for current events like meetings.

In the end, we chose **Plan A** because of its **comprehensiveness**, **flexibility** and **sustainability**. The program covers three high-frequency life scenarios: diet, decision making, and time management. From the **user's point of view**, an **integrated app** is a better choice, and fragmented software is too inconvenient to use. And through the collection of various data, users can have a **personal customized AI butler**, rather than a rigid question-and-answer robot.

## Evaluation Impact

#### Evaluation method:

As for the evaluation method, we believe that it is reasonable to adopt a mixed framework to evaluate from both the technical and user levels:

- **Technical Validation:** Algorithm accuracy tests (e.g., 95% food recognition precision in diverse lighting conditions).
- **User Studies:** 3-month beta trials with 500+ participants, tracking metrics like task completion rates and user retention.

For data collection, **audio recordings and transcripts, and Questionnaire** data is saved.

#### Key findings:

**User Trust Challenges:** 28.5% expressed initial privacy concerns, mitigated by transparent data control features added post-trial.

#### Human-Centric Impact:

- **Empowered Health Choices:**

Simplified nutrition analysis enabled time-constrained users to make informed dietary decisions faster.

- **Reduced Decision Fatigue:**

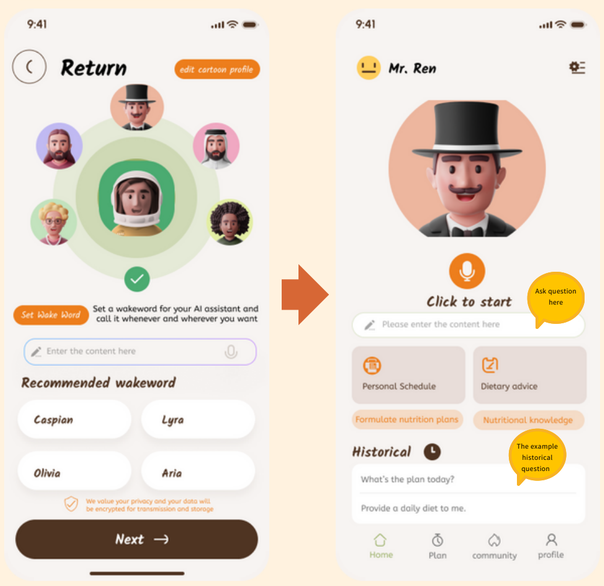
Smart scheduling reduced daily planning time by 40%, freeing mental bandwidth for creative tasks.

The **evaluation** takes part in the each **iteration** of our interaction design. Just as the following, shows how the functionalities realized and improved based on the evaluation through these two methods.

| | | | |
|---|---|---|---|
| Iteration 1 | Users want to personalize the appearance of the virtual butler | Plan A: Added DIY feature to allow customization of the butler's image and add wake word.<br>Plan B: Provide a variety of styles of the virtual image template, users can choose gender, hairstyle, clothing, skin color and other basic attributes. | We choose **Plan A**:Letting users completely customize their AI butler in a way that makes it feel more personal and unique is cooler than picking from boring templates. |
| Iteration 2 | Users are worried about how their data is used | Added privacy disclaimers and on-screen reminders explaining data usage policy | |
| Iteration 3 | Users want more than reminders; they want contextual suggestions like dining/hotel options | Plan A: Recommend highly matched restaurants/hotels in real time based on location, preference and scene, and show why.Enter requirements through the dialog box to dynamically generate personalized recommendations<br>Plan B: Proactively push selected recommendations that fit the user's habits at the right time | We chose **PlanA** over PlanB because it provides smarter, real-time personalized recommendations based on the user's actual needs and preferences |
| Iteration 4 | It is difficult for users to understand the nutritional analysis of the food they are consuming | Plan A: The peripheral-body camera scans the food and nutrients the user is digesting in real time, and displays missing nutrients and foods<br>Plan B: Visual presentation of nutrition data in colors/graphs and provide health scores | Choose **Plan A** because it not only shows your nutrition gaps but actively tells you which foods to eat to fix them, while Plan B just displays data without solutions. |




## Decision Support




## Smart Eating Assistant







## Scheduling System




## Final Product

## Conclusion

### AI-Powered smart assistant: Integration and personalization

This project has successfully developed an AI-powered smart assistant system integrating intelligent decision analysis, schedule management, and dietary recommendations, achieving innovation in vision-based quantitative nutrient analysis through photographic data.

### Future improvement:

#### Cross-Platform Ecosystem Integration

Seamlessly sync with smart home devices, wearables (e.g., fitness trackers), and vehicles to create a unified life-management ecosystem. For example, automatically adjust home lighting based on stress levels detected via voice tone analysis.

#### Augmented Reality (AR) Interfaces

Enable AR-guided meal prep (e.g., holographic cooking instructions with calorie adjustments).
Visualize daily schedules as interactive 3D timelines in mixed-reality environments.